COMPUTER NETWORKS AND INTERNET APPLICATIONS

Modules 4 and 5

High-yield notes, comparisons, mechanisms, and question bank

Exam Focus.

Study strategy. Build each long answer around a definition, a labelled diagram or flow, working steps, benefits and limitations, and a concrete example. Give particular attention to symmetric versus asymmetric cryptography, firewall classes, VPN types, TLS versus IPsec, Internet components and routing, cloud service models, and CDN operation. The supplied slides do not specify an exam pattern or marks distribution; use the question bank as revision prompts, not as a prediction.

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Module 4: Introduction to Network Security

Security goals, threats, and mechanisms

Network security comprises policies, technologies, protocols, and practices that protect networks and transmitted data from unauthorized access, disclosure, alteration, misuse, and disruption. The central goals are confidentiality (prevent reading by unauthorized parties), integrity (detect unauthorized changes), authentication (verify identity or origin), and availability (keep services usable). Digital signatures can also provide evidence supporting non-repudiation.

Threat	Meaning	Example
Eavesdropping	Unauthorized observation of traffic	Reading unencrypted public Wi-Fi messages
Spoofing	Falsifying identity or address	A fake site impersonating a bank
Malware	Software that damages, monitors, or controls systems	Ransomware spreading across a network
MITM	Attacker intercepts or relays communication	Interception of an insecure connection
DoS	Exhausting resources to deny service	Flooding a server with requests
Phishing	Deceiving users into revealing information	Fraudulent credential email
Data modification	Unauthorized alteration in transit	Changing a transaction amount

Three complementary controls recur throughout this module: cryptography protects information, firewalls control permitted traffic, and VPNs create protected communication paths over untrusted networks.

Cryptography

Cryptography uses mathematical algorithms to protect information. Plaintext is readable input; encryption transforms it to ciphertext using a cipher and key; decryption recovers plaintext. Cryptanalysis studies attacks on protections, while cryptology encompasses cryptography and cryptanalysis.

Symmetric and asymmetric systems

Feature	Symmetric key	Asymmetric key
Keys	Same secret (or closely related key material) encrypts and decrypts	A public/private key pair; publish public key, protect private key
Strength	Efficient for bulk data	Supports public-key authentication and key establishment
Challenge	Securely share and protect the secret	More computationally intensive; bind public key to correct owner
Examples	AES, ChaCha20; DES and 3DES are obsolete/legacy for modern use	RSA, ECC; ECDH and Diffie–Hellman establish shared secrets

Symmetric flow: establish shared key → encrypt plaintext → transmit ciphertext → decrypt with shared key. Asymmetric operations depend on purpose: public-key encryption is typically addressed to the recipient's public key and opened with the private key; a digital signature is made with the sender's private key and verified with the corresponding public key.

Do Not Confuse.

> A hash is not encryption: it is designed to be one-way and has no decryption step. A digital signature is not simply encryption of the whole message. Key ownership and correct key management are essential.

Hashes, integrity, and signatures

A cryptographic hash maps arbitrary-length input to a fixed-length digest. SHA-256 and SHA-3 are examples; MD5 and SHA-1 are unsuitable where modern collision resistance is required. To check

integrity, the receiver hashes the received message and compares the result with a digest delivered through a trusted mechanism. A digest sent alongside data over the same untrusted channel alone does not prove authenticity. A digital signature signs a message digest with a signing key; the receiver verifies it using the corresponding public key. It supports integrity, origin authentication, and evidence useful for non-repudiation, provided the key is reliably associated with its owner.

Firewalls

A firewall is hardware, software, or both, that monitors and controls inbound and outbound traffic according to policy. It defines a boundary between security zones and may evaluate addresses, ports, protocols, connection state, and application data. A common policy permits required traffic and denies other traffic by default.

Type	How it works	Strength / limitation
Packet filtering	Checks packet headers such as source/destination IP, ports, protocol, direction	Fast, simple; stateless rules lack session context and application insight
Stateful inspection	Tracks sessions in a connection table and judges packets in connection context	Allows expected return traffic; consumes state and does not detect every malicious application action
Application proxy	Terminates/intermediates connections and examines application requests	Detailed application policy; may add latency and needs application support

Firewalls reduce exposure, restrict services, segment networks, and can record selected events. They do not by themselves stop phishing, insider misuse, malware using allowed channels, vulnerable permitted services, or every attack in encrypted traffic. Use them with endpoint security, strong authentication, patching, monitoring, backups, and user awareness.

Virtual Private Networks

A VPN establishes a protected logical connection over an existing network. Depending on its protocol and configuration, it can provide tunnelling, encryption, integrity, and authentication. Typical sequence: client/gateway connect; authenticate; establish keys and parameters; encapsulate and protect traffic; carry it over the public network; verify and decapsulate at the gateway; enforce access and routing policy.

- **Remote access:** one user/device securely reaches a private network (e.g., staff connecting from home).
- **Site-to-site:** gateways connect networks, often branch offices; intranet joins sites in one organization, extranet joins authorized organizations.
- **Host-to-host:** a protected link joins two specific computers or servers.

A VPN protects a channel when correctly configured; it does not make all destinations or applications inherently safe. Firewall policy, endpoint health, authentication, routing, and DNS configuration still matter.

Remember.

> College example: the user authenticates to remote-access VPN; cryptography protects the tunnel; the firewall permits only authorized access to the server; the server authenticates and authorizes the user; monitoring records events. Each control has a distinct role.

Security protocols: TLS and IPsec

Security protocols specify how parties authenticate, establish keys, and protect data. Typical objectives include confidentiality, integrity, authentication, key establishment, and anti-replay protection.

	TLS	IPsec
Layer / scope	Between application protocol and transport; secures application connections such as HTTPS	IP layer; protects IP packets between hosts or security gateways
Common use	Web, APIs, and protected application protocols	VPNs, especially site-to-site connections
Main elements	Handshake negotiates parameters/keys and authenticates; record protection carries application data	Security policies and Security Associations (SAs); AH or ESP; transport or tunnel mode
Notes	SSL is obsolete; modern deployments use TLS. TLS 1.3 differs from legacy diagrams	Tunnel mode can protect an entire inner IP packet; transport mode protects payload while retaining outer IP header

TLS 1.3 simplified flow: ClientHello with supported capabilities and key share → ServerHello and selected parameters → derive shared traffic secrets → server proves identity (usually certificate and private-key proof) → Finished messages verify handshake transcript → encrypted application data. Certificate checks include chain, hostname, and validity. IPsec applies policy to traffic, selects an SA with algorithms and keys, protects packets with Encapsulating Security Payload (ESP) or Authentication Header (AH), and the receiver checks integrity/anti-replay and decrypts when ESP is used. ESP can provide confidentiality, integrity, and authentication; AH provides integrity/authentication but not confidentiality. Exact services depend on configuration.

Do Not Confuse.

SSL is historical and obsolete. TLS protects application connections; IPsec protects at the IP layer. AH does not encrypt. ESP and AH are IPsec protocols, not VPN types.

Module 5: Structure and Functioning of the Internet

Internet structure and packet delivery

The Internet is a decentralized global interconnection of networks that communicate primarily using TCP/IP. It uses packet switching, standard protocols, and multiple paths; networks and devices may differ in hardware and operating systems. Main components include end systems, access networks, routers, ISPs, backbone links, data centres, and interconnection points.

Host / client ↔ access network ↔ access ISP ↔ regional / transit networks and IXPs ↔ destination ISP ↔ serv

Hosts (laptops, phones, servers, sensors) run applications at the network edge. Access technologies include Ethernet, Wi-Fi, fibre-to-the-home, DSL, cable, cellular, and satellite. Routers forward IP packets: receive packet → inspect destination IP → consult forwarding table → choose next hop/interface → forward. Routing protocols help populate routes; individual packets may follow different paths.

ISPs, backbone, and Internet Exchange Points

An ISP provides connectivity to customers and networks. Common roles include access ISPs, regional aggregators, transit providers, backbone operators, and content/cloud network operators; modern interconnection is not a rigid hierarchy. ISPs may provide access, IP configuration/address delegation, routing, DNS resolvers, traffic management, security, support, and interconnection. The backbone is high-capacity long-distance infrastructure: fibre links, high-capacity routers, transmission systems, and interconnections among operators. An Internet Exchange Point (IXP) provides shared switching infrastructure where networks can exchange traffic directly. Participants connect routers, establish agreements and routing (often BGP), and exchange frames across the switching fabric. An IXP is generally not itself the transit provider carrying all traffic between participants. Direct exchange can reduce path length, latency, and transit cost.

Autonomous Systems and routing

An Autonomous System (AS) is a set of IP networks and routers under one administration or common routing policy. A public AS has an ASN. ASes organize inter-domain routing while allowing each organization to manage its internal routing and advertise reachable prefixes.

- **Stub AS:** connects to outside networks but does not normally carry transit traffic between unrelated networks.
- **Multihomed AS:** connects to two or more external networks for resilience or policy choice; this alone does not imply transit.
- **Transit AS:** carries traffic between other networks, as an ISP may do.

An Interior Gateway Protocol (IGP) routes within an AS; examples include RIP, OSPF, and IS-IS. An Exterior Gateway Protocol exchanges reachability between ASes; BGP is the principal inter-domain protocol used on today's Internet.

Do Not Confuse.

> An AS is an administrative/routing domain, not one building. An IXP is an interconnection facility, not synonymous with an ISP. IGP is internal; BGP handles inter-domain reachability.

Cloud computing

Cloud computing offers on-demand network access to a shared pool of configurable resources (servers, storage, networks, applications) that can be provisioned and released with limited provider interaction. The front end includes client devices, browsers, apps, and dashboards; the back end includes physical servers, virtual machines/containers, storage, databases, network, virtualization, management, and security.

Model	Provider supplies	Customer typically manages
IaaS	Virtual machines, storage, networking	Operating system, deployed applications, much configuration
PaaS	Managed runtime/platform to build and deploy applications	Application code and data
SaaS	Complete network-accessible application	Use/configuration of the application

Five commonly taught characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service. Deployment models: public (provider-operated and shared with logical isolation), private (dedicated to one organization), hybrid (connected distinct environments), and community (shared by organizations with common needs).

Content Delivery Networks

A CDN distributes servers and network infrastructure geographically so users can receive eligible content from an appropriate edge location instead of always fetching it from one origin. It can reduce latency and origin load, improve scalability and availability, and distribute media efficiently. Typical content includes images, scripts, stylesheets, downloads, video, and selected API responses.

1. The client requests a resource.
2. DNS or another traffic-steering mechanism directs it to a suitable CDN location/PoP.
3. The edge checks its cache and rules.
4. On a cache hit, it serves a valid copy; on a miss, it fetches from origin or an upstream cache.
5. It delivers the response and may retain an eligible copy according to cache controls/TTL.

Origin stores authoritative content; edge servers cache and deliver content; a Point of Presence (PoP) is a network location with provider equipment. Selection depends on routing, load, availability, and policy, not simply geographic distance. Pull CDNs fetch on first request; push CDNs receive pre-distributed content. Static content is often cacheable; dynamic/personalized data needs careful policy. Streaming

often uses successive segments and adaptive bitrate. CDN security options may include DDoS mitigation, WAF, TLS management, and rate limits, but do not replace application and origin security.

Do Not Confuse.

A cache hit means a usable cached object was found; a miss means the edge must obtain it elsewhere. Not all dynamic content should be cached: personalized or sensitive responses require strict access and cache controls.

High-yield comparisons

Pair	First	Second
Encryption vs hashing	Reversible with appropriate key; confidentiality	One-way digest; integrity comparison
VPN vs firewall	Creates protected connectivity/tunnel	Permits or blocks traffic by policy
TLS vs IPsec	Application connection protection	IP-layer packet protection
Cloud vs CDN	On-demand compute/storage/platform/application	Distributed content delivery and caching
Pull vs push CDN	Edge fetches content on demand	Content distributed to edge in advance

Answer skeletons for 10-mark questions

Explain cryptographic techniques and digital signatures [10 marks]

Answer Skeleton.

1. Define cryptography and goals.
2. Explain plaintext, ciphertext, cipher, key, encryption/decryption.
3. Compare symmetric and asymmetric methods with examples.
4. Describe hashing and integrity checking.
5. Give signature creation and verification steps.
6. Add uses and key-management caveat; conclude with how mechanisms complement one another.

Explain firewall types and VPN operation [10 marks]

Answer Skeleton.

Define firewall and policy; compare packet filter, stateful inspection, and proxy; state strengths and limits; define VPN and outline authentication, key setup, encapsulation/protection, transit, verification, and forwarding; distinguish remote-access, site-to-site, and host-to-host; give college example and note layered security.

Describe TLS and IPsec and compare them [10 marks]

Answer Skeleton.

Define security protocol and goals; explain TLS position and handshake/data flow, certificate validation, and SSL obsolescence; explain IPsec position, policy, SA, ESP/AH and modes; compare layer, scope, and common use; draw labelled flows.

Explain Internet structure, ASes, and routing [10 marks]

Answer Skeleton.

Define Internet and packet switching; draw host-to-access-to-ISP-to-backbone/interconnection-to-server path; explain router forwarding; describe ISP and IXP; define AS/ASN and stub, multihomed, transit; compare IGP and BGP.

Explain cloud models and CDN operation [10 marks]

Answer Skeleton.

Define cloud and list five characteristics; explain front/back end; compare IaaS, PaaS, SaaS and public/private/hybrid/community; define CDN, origin, edge, PoP; narrate request, steering, hit/miss, origin fetch and delivery; discuss benefits, cache controls, and security limits.

Question bank

Definitions / short answers

1. Define confidentiality, integrity, authentication, and availability.
2. Define plaintext, ciphertext, key, cipher, and digest.
3. What is an Autonomous System and ASN?
4. Define IXP, PoP, cache hit, and cache miss.
5. What is a stateful firewall?

5-mark questions

1. Compare symmetric and asymmetric cryptography.
2. Explain hash-based integrity checking and digital signatures.
3. Describe firewall types and their limitations.
4. Differentiate remote-access and site-to-site VPNs.
5. Compare IaaS, PaaS, and SaaS.
6. Explain how an IXP supports interconnection.
7. Describe CDN cache-hit and cache-miss processing.

10-mark questions

1. Explain cryptography, its goals, types, hashes, and digital signatures.
2. Describe firewall architecture, classes, advantages, and limitations.
3. Explain VPN operation and the relationship between VPNs, cryptography, and firewalls.
4. Compare TLS and IPsec, with handshake/packet flows.
5. Describe Internet structure and the role of ISPs, routers, IXPs, and ASes.
6. Explain cloud computing characteristics, service models, and deployment models.
7. Explain CDN architecture, operation, delivery types, and benefits.

Last-minute revision

SECURITY GOALS
Confidentiality, integrity, authentication, availability; signatures support evidence of origin.

CRYPTO
Symmetric: shared secret, fast bulk protection. Asymmetric: public/private pair. Hash: one-way digest. Signature: private-key signing, public-key verification.

FIREWALL
Packet header filter; stateful tracks sessions; proxy inspects application requests.

VPN
Authenticate → establish keys → encapsulate/protect → tunnel → verify/forward.

PROTOCOL LAYERS
TLS: application connection. IPsec: IP layer. SSL obsolete. ESP can encrypt; AH does not.

INTERNET
Host → access → ISP → interconnection/backbone → destination. Router forwards by IP and table.

ROUTING
IGP within AS; BGP between ASes. IXP enables direct exchange.

CLOUD
IaaS infrastructure; PaaS application platform;

SaaS complete application.

CDN

Client → edge/PoP → cache hit or origin fetch → delivery.

Source scope

These notes synthesize the supplied Module 4 slides, *Introduction to Network Security: Cryptography, Firewalls, and VPNs*, and Module 5 slides, *The Structure and Functioning of the Internet*. Terminology and emphasis follow those materials; the slides do not state an exam pattern or marks allocation.

Study principle: understand the mechanism first, then memorize the exam wording.